

Lab 6 - Cron Jobs and Heartbeat

Schedule recurring agent tasks with cron jobs and monitor agent uptime with heartbeat.

Lab environment: Hostinger VPS (from Lab 1) or Docker Desktop (Windows 10/11, macOS 12+, Ubuntu 22.04+)

Prerequisite: Lab 3 completed - at least one channel active. Estimated time: 30 minutes

Docker Desktop users: Prefix every openclaw command with docker exec -it openclaw.

Example: docker exec -it openclaw openclaw cron list

Cron vs Heartbeat

Feature	Cron Jobs	Heartbeat
Purpose	Run a task on a schedule	Confirm the agent is alive
Trigger	Time-based (e.g. every morning)	Interval ping
Output	Task result sent to channel	Short status message
Use case	Daily reports, weekly research	Uptime monitoring

Part A - Cron Jobs

Step A1 - Create a Daily Morning Briefing

```
# VPS
openclaw cron add \
  --cron "0 9 * * *" \
  --name morning-briefing \
  --channel telegram \
  "Summarise today's top AI news"

# Docker Desktop
docker exec -it openclaw openclaw cron add \
  --cron "0 9 * * *" \
  --name morning-briefing \
  --channel telegram \
  "Summarise today's top AI news"
```

Cron syntax: minute hour day month weekday 0 9 * * * = every day at 9:00 AM

Step A2 - Create a Weekly Research Cron

```
# VPS
openclaw cron add \
  --cron "0 8 * * 1" \
  --name weekly-ai-news \
  --channel telegram \
  "Research top AI agent news this week and send a summary"

# Docker Desktop
docker exec -it openclaw openclaw cron add \
  --cron "0 8 * * 1" \
  --name weekly-ai-news \
  --channel telegram \
  "Research top AI agent news this week and send a summary"
```

0 8 * * 1 = every Monday at 8:00 AM

Step A3 - List All Cron Jobs

```
# VPS
openclaw cron list

# Docker Desktop
docker exec -it openclaw openclaw cron list
```

Expected output: `` NAME SCHEDULE CHANNEL STATUS morning-briefing 0 9 * * * telegram enabled
weekly-ai-news 0 8 * * 1 telegram enabled ``

Step A4 - Run a Cron Immediately (Test)

```
# VPS
openclaw cron run morning-briefing

# Docker Desktop
docker exec -it openclaw openclaw cron run morning-briefing
```

Expected: Telegram receives the briefing without waiting for the schedule.

Step A5 - Disable and Enable a Cron

```
# VPS
openclaw cron disable morning-briefing
openclaw cron list
openclaw cron enable morning-briefing

# Docker Desktop
docker exec -it openclaw openclaw cron disable morning-briefing
docker exec -it openclaw openclaw cron enable morning-briefing
```

Step A6 - Delete a Cron

```
# VPS
openclaw cron rm weekly-ai-news

# Docker Desktop
docker exec -it openclaw openclaw cron rm weekly-ai-news
```

Part B - Heartbeat

Heartbeat sends a regular ping to confirm the agent and gateway are running.

Step B1 - Enable Heartbeat

```
# VPS
openclaw system heartbeat enable

# Docker Desktop
docker exec -it openclaw openclaw system heartbeat enable
```

Step B2 - Check Last Heartbeat

```
# VPS
openclaw system heartbeat last

# Docker Desktop
docker exec -it openclaw openclaw system heartbeat last
```

Step B3 - Disable Heartbeat

```
# VPS
openclaw system heartbeat disable

# Docker Desktop
docker exec -it openclaw openclaw system heartbeat disable
```

Verification

Check	Expected
openclaw cron list	morning-briefing shown as enabled
openclaw cron run morning-briefing	Telegram receives the briefing immediately
openclaw system heartbeat last	Last heartbeat timestamp shown

Troubleshooting

Symptom	Fix
Cron does not fire on schedule	Gateway must be running continuously - check openclaw gateway status

Symptom	Fix
cron run returns error	Verify channel is active: openclaw channels status
Docker: cron fires but no Telegram message	Confirm bot token and pairing are still valid
cron rm not found	Command is rm not delete

Reference

- Cron docs: <https://docs.openclaw.ai/cron>
- Heartbeat: <https://docs.openclaw.ai/heartbeat>